EXPERIMENT NO : 1a

ANALYSE THE TREND OF DATA SCIENCE JOB POSTINGS OVER THE LAST DECADE

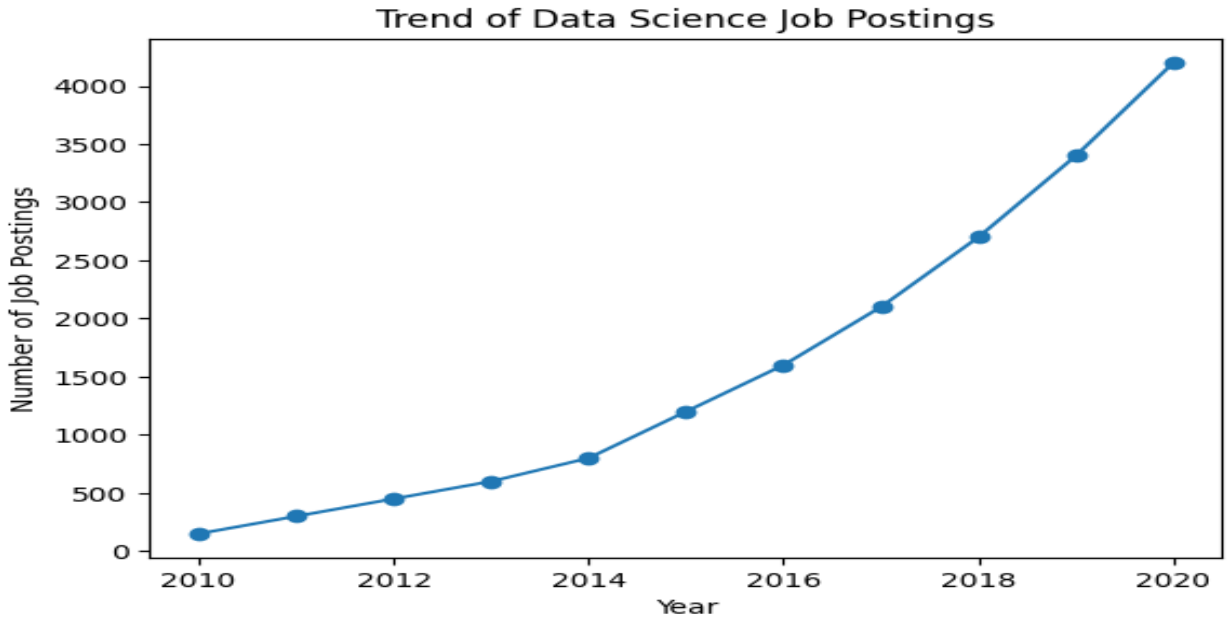
AIM :

To analyze the trend of data science job postings over the last decade.

CODE :

```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
    'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

OUTPUT :



RESULT :

Thus,the given data was analyzed successfully and required output was obtained.

EXPERIMENT NO : 1b

ANALYZE AND VISUALIZE THE DISTRIBUTION OF VARIOUS DATA SCIENCE ROLES(DATA ANALYST,DATA ENGINEER,DATA SCIENTIST,ETC....) FROM A DATASET

AIM :

To analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

CODE :

```
Import pandas as pd

import matplotlib.pyplot as plt

roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer', 'Business Analyst']

counts = [300, 500, 450, 200, 150]

plt.bar(roles, counts)

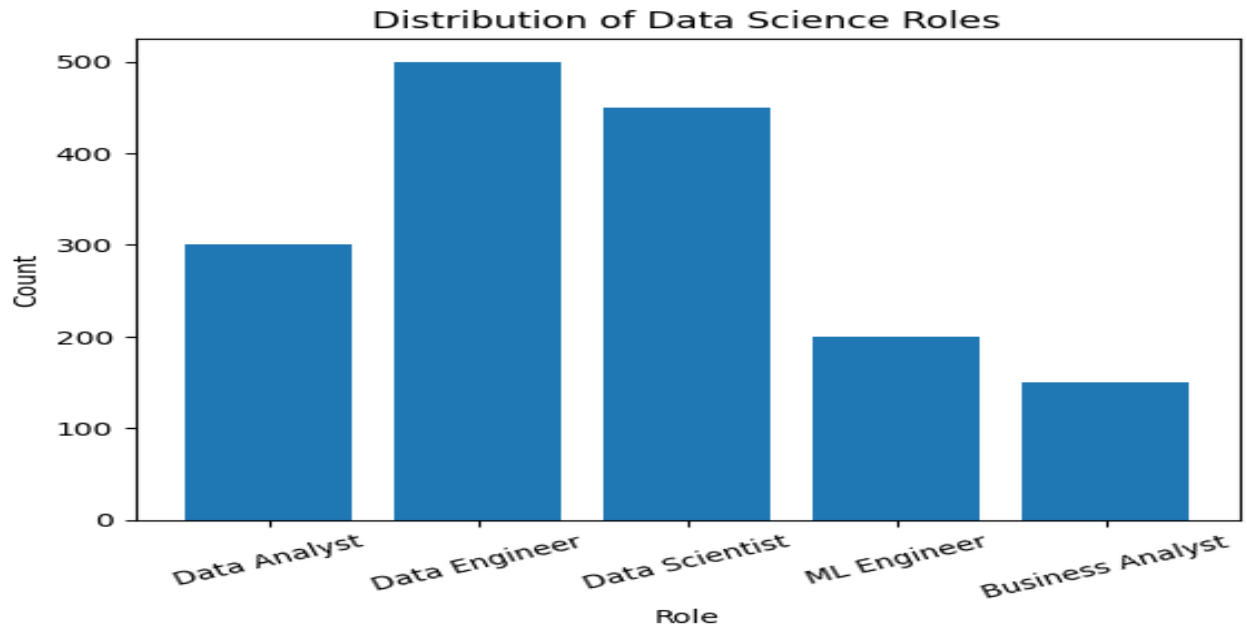
plt.title('Distribution of Data Science Roles')

plt.xlabel('Role')

plt.ylabel('Count')

plt.show()
```

OUTPUT :



RESULT :

Thus,the given data was analyzed successfully and required output was obtained.

EXPERIMENT NO : 1c

CONDUCT AN EXPERIMENT TO DIFFERENTIATE STRUCTURED, UN-STRUCTURED AND SEMI-STRUCTURED DATA BASED ON DATA SETS GIVEN

AIM :

To conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

CODE :

```
import pandas as pd

structured_data = pd.DataFrame({

  'ID': [1, 2, 3],

  'Name': ['Alice', 'Bob', 'Charlie'],

  'Age': [25, 30, 35]

})

print("Structured Data:\n", structured_data)

unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image, or a video file."

print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {

  'ID': 1,

  'Name': 'Alice',

  'Attributes': {
     'Height': 165,

    'Weight': 68}}

print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT :

```
Structured Data:

  ID    Name  Age

0  1   Alice   25

1  2     Bob   30

2  3 Charlie   35
```

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}

RESULT :

Thus,the given data was analyzed successfully and required output was obtained.

EXPERIMENT NO : 1d

CONDUCT AN EXPERIMENT TO ENCRYPT AND DECRYPT GIVEN SENSITIVE DATA

AIM :

To conduct an experiment to encrypt and decrypt given sensitive data.

CODE :

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

f = Fernet(key)

token = f.encrypt(b"Raj alakshmi Engineering College")

print(token)

print(f.decrypt(token))

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Rajalakshmi Engineering College."

cipher_text = cipher_suite.encrypt(plain_text)

decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

OUTPUT :

```
Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABqit0q7k64s3x0ghEn7XssbvJ7TcfLgH7LStnEp-
lX0sArwQDFdByI9sMd5NrfCwUdyX3NFPNRzdszT_adRhOfefzKLkFHtZuuwJpZ6B6J_0KEPFE='

Decrypted Data: b'Rajalakshmi Engineering College'
```

RESULT :

Thus,the given data was analyzed successfully and required output was obtained.